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Virtual Commissioning – Scientific review and exploratory use cases in advanced production systems

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Abstract

Virtual Commissioning increases quality and efficiency in production engineering whilst also decreasing required time. Even though it is a foundation for state-of-the-art production systems, it is not in widespread use yet. In this contribution, an overview of current scientific approaches and use cases for Virtual Commissioning is given. Furthermore, additional fields of interest are derived and detailed on exploratory use cases. The findings show that Virtual Commissioning is a crucial factor to engineering of advanced, sustainable production systems as well as a cornerstone of the Digital Twin.

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1. Introduction

Currently many companies still engineer their production systems in the established, non-synchronized way. This means that the engineering results of the involved disciplines, namely mechanical, electric and software engineering, most of the time collide at the commissioning for the first time. This results in delays during the commissioning process due to necessary adjustments in the different disciplines [1]. As shorter product lifecycles lead to more frequent changes affecting production systems and lines, manufacturers have to repeat the process of commissioning more and more frequently [2]. To speed up the commissioning process, it is possible to virtually build up production systems in advance and commission them in a simulation environment. This method is called Virtual Commissioning (VC) [3].

At first, this paper defines the term VC and shows the state of the art in VC. Furthermore, it gives an overview of current applications in the field of VC. Moreover, different exploratory

use cases are presented. In front of this background, the importance of VC for the Digital Twin (DT) of machines is explained. Finally, a brief summary and an outlook on further work are given.

2. Scientific review

VC is defined as the early development and validation of programmable logic controller (PLC) code using a simulation model [1]. The term VC appears in a wide range of applications. Starting from VC of logistic systems, the term is used in the context of process simulation and automation system validation as well [4–6]. In this paper, VC focusses on the optimization and validation of the latter. The code of the PLC controls a virtual model, also called behavior model, which behaves like the real machine. Thus, the PLC program can be optimized and validated before real commissioning [7]. The fact that the software components are becoming more

important strengthens the application of the described type of VC [1].

Fig. 1 shows the difference between the engineering process with and without VC. First, the production system has to be modelled. At best, this task starts during the engineering process, which enables the model to detect inconsistencies and errors in the construction [8]. Due to the fact, that VC can be done parallel to the production and assembly process, valuable time in the pending commissioning process can be saved by optimization loops during those processes outlined by the arrows in Fig. 1. Furthermore, VC enables a safer commissioning since previous detected errors are solved in prior to the real commissioning. [9]

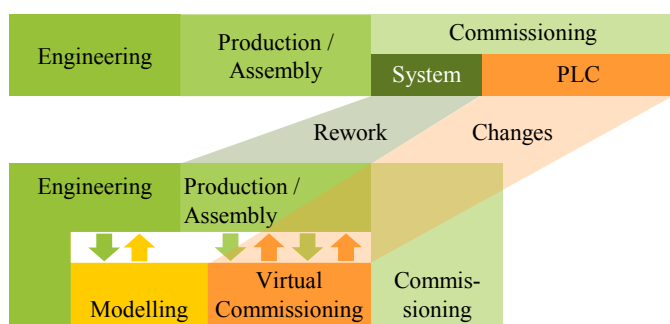


Fig. 1. Engineering with and without VC [3].

There are different approaches for VC of production systems. Fig. 2 shows the basic differences of VC setups. All VC processes base on a virtual model, which is linked with a PLC. In case of a hardware in the loop (HiL) simulation, the PLC is a real hardware controller [10]. Consequently, it is possible to perform the VC with the PLC that is integrated in the production system afterwards. On the other hand, VC can be applied on an emulated PLC, which is called software in the loop (SiL) simulation. Therefore, no hardware PLC is required, which is one of the main advantages of SiL simulations. Furthermore, it is possible to do a reality in the loop simulation, combining an emulated PLC with the production system to test certain real components [11]. Regarding the virtual model, it is possible to combine it with components of the real machine to test their future function, which results in a hybrid simulation.

Furthermore, it is possible to perform VC in different software tools. All of them have their advantages regarding performance, reliability and usability. A precise comparison of the available tools is a future field of action.

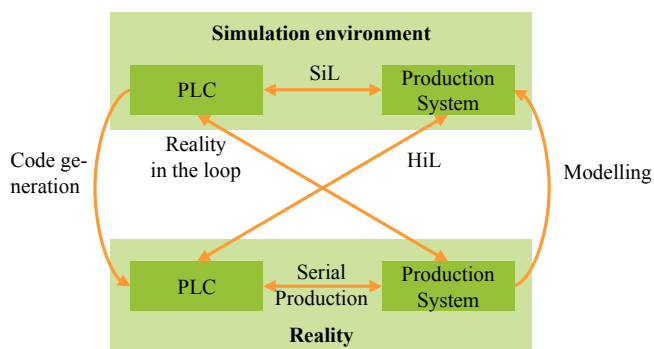


Fig. 2. Different concepts for VC [11,12].

The introduction of VC in existing engineering processes is difficult because of various reasons. The main issues are the high implementation costs at the beginning as well as the considerable complexity of VC in general. [13]. The initial modelling process causes most of those difficulties [11]. For the optimization of the modelling, different approaches exist. Firstly, current research focusses on the automated generation of VC models [14]. Thereby, the appropriate level of abstraction regarding the simulation model is evaluated [15]. Apart from that, the reutilization of VC models benefits from the advancing standardization of production systems components [16].

With the finished behavior model, basic kinematics of the production system can be checked on collisions and bad designs. The automation solution in the form of the PLC code can be developed using the behavior model in a simulation environment. The first goal is to validate the manual functions of the production system. Afterwards, the automation mode as well as the special features such as special operating modes as well as setting-up processes are validated. One main advantage of VC is the possibility to verify the planned cycle time of the production system [17]. Furthermore, scenarios, which can result in costly crashes in reality, can be tested. These tests can be performed by manipulating the simulation environment manually. Current research focusses on the automated creation of test cases generated by the structure of the production systems. [18]

Table 1 shows a representative extract of the latest research regarding the fields of VC application. The levels of abstraction regarding VC separate the rows into the automation, robotics, process and energy level. The columns show the different fields of application of VC, divided in logistics, assembly processes, manufacturing processes and test environments.

Table 1. Extract of current VC application sorted by level of abstraction (rows) and field of application (column).

	Logistics	Assembly	Manufacturing	Test bed
Automation	[19][20] [21][22]	[7][11] [23][24] [25][26]	[1][5]	[4][16][27] [28][29][30] [10]
Robotics		[6][13] [31][32] [33]	[34][35]	[36]
Process			[37]	[38]
Energy		[23][39]		

Most applications of VC focus on the automation level, which bases on the optimization and validation of the PLC code. Thereby, most of the applications show functionalities by using demonstrative test beds. Moreover, VC on the automation level is often deployed in the field of assembly processes regarding the automation level. Logistics applications only appear on the level of automation. In the fields of robotics, there are several applications, which mainly focus on the assembly regarding handling operations. The rare use cases concerning processes are in the manufacturing as well

as in test environments. It shows that the method of VC does not focus the process as much as the automation level, because VC cannot contribute on the process to the same extent as on the automation solution. Another reason for the small amount of applications on the process level is the conflict between the real-time capability and the required processing power for the process simulation. VC with focus on the energy simulation of production systems currently only exist in the field of assembly. Overall, the provided examples show that simulation in form of VC is used in a wide area of application, though the potential of VC is not fully exploited. [13]

3. Exploratory use cases

Hereafter, three different exploratory use cases are presented. At first, the use case of “human-in-the-loop” commissioning shows the special features regarding the interaction between humans and robots as well as the corresponding safety regulations. This use case can be located at the robotics level in the field of assembly shown in Table 1. The next use case focusses on the growing importance of environmentally friendly production by targeting the VC of smart grids, which are driven by a DC power supply for an appropriate integration of renewable energy sources on the grid level. Latter belongs to the energy level in the application of assembly. The last use case describes the features of the special machine construction (SMC) in VC of the PLC, concerning the limited reutilization of simulation models. Therefore, it is localized in the field of assembly on the automation level.

3.1. Human robot interaction VC

The field of hybrid production systems, especially focusing on human-robot cooperation (HRC), has gained attention in the last years. However, as there is no physical separation between human and robot as well as contact may – or is intended to – take place, a thorough risk evaluation and promotion of worker acceptance become major factors for implementation success. To avoid costly prototype assemblies and iterative changeovers, a VC of such systems can be used to identify and avert risky system design or behavior as well as to introduce workers to their “new colleague” and its behavior in a safe, virtual space. As collaborative applications are often retrofit applications to a productive manual system for economic or ergonomic reasons, the changeover time has to be as short as possible. VC is hence a key for the successful implementation of HRC. Kinematics and human simulation can already provide insights into the ergonomic and performance-related aspects of the designed hybrid production system. [24]

To further achieve a risk assessment or virtual training, VC has to be used and extended by a realistic robot controller simulation as well as a human simulation, not only focusing on planned human behavior but also encompassing foreseeable misuse and accidents. This may require a Motion Capturing (MC) system and, to enable realistic human immersion and feedback, the real-time visualization via a Virtual Reality (VR) head-mounted display. [32] An architecture detailing the

requirements of HRC VC compared to classic VC is shown in Fig. 3. As safety evaluation is a definitive goal of the use case, another key property of “human-in-the-loop” commissioning is the integration of safety features into the simulation. It may hence be necessary to include physical HiL.

Additional to the application in production system engineering, the proposed method is especially useful for production system optimization or reconfiguration. As this usually includes changes to the hardware, the PLC or the robot controller, it also makes a recertification of the human-robot system necessary. To reduce or eliminate the consequential system downtime, VC offers the ability to evaluate the changes virtually and enable employee training without production throttling. As hybrid production systems are mostly deployed due to their high flexibility in comparison to common automated systems, this aspect is of high importance.

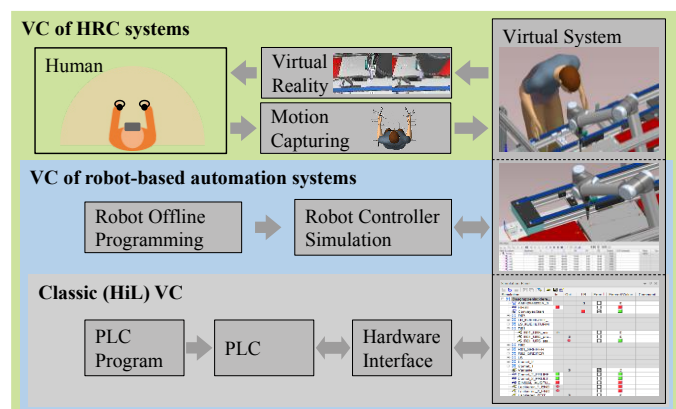


Fig. 3. Additional aspects of HRC system VC.

3.2. VC of smart grids

The European Union (EU) aims to reduce greenhouse gas emissions to at least 40 % by 2030 and increase the share of renewables to 27 % [40]. Consequently, the EU member states must take action in order to achieve this goal. Thereby, one challenge is the volatility of many renewable energy sources. In the future, it will not be sufficient just to adapt production within industrial plants to the changed industrial framework conditions with regard to energy supply. Instead, it will be necessary to pay special attention to the development and dimensioning of the production infrastructure, since the necessary energy flexibility can only be achieved through targeted energy management. In addition, it turns out that this is a complex interaction of consumers, storages and producers of electrical energy.

As part of VC, process management tests are already state of the art. In order to be able to test and optimize the energy management equally profitably in the run-up, an exact and detailed forecast of the energy requirements of the individual consumers (condition-based energy data) is necessary. Determining this in advance of the VC is a challenge that must be posed as a request to the component or production plant manufacturer. Another possibility is to determine the

consumption values of the components in a different composition and subsequently transfer them. [39]

If, in addition to the consumers, the generators and storages of electrical energy shall be regarded, nowadays a second control for the management of energy is essential. This should also be tested virtually before the run-up. For the holistic consideration of the virtual commissioning, the communication between the two controllers is the major challenge. Here, different types of connections are possible. Fig. 4 shows the tools that are currently tested for VC including the energy simulation. The overall structure consists of the energy and process management as well as the programming environment for the PLC. The latter takes place in the tool called Totally Integrated Automation Portal (TIA Portal) [41]. The energy management can be performed in Plant Simulation or Amesim [42,43]. For VC the tool SIMIT is used [44]. The different levels of processes are mapped in the tools Process Simulate, Plant Simulation and the Mechatronic Concept Designer [45,46]. All the tools originate from the Siemens toolchain, which enables a comparatively easy connection between the tools using the PLC and the Open Platform Communications Unified Architecture (OPC UA).

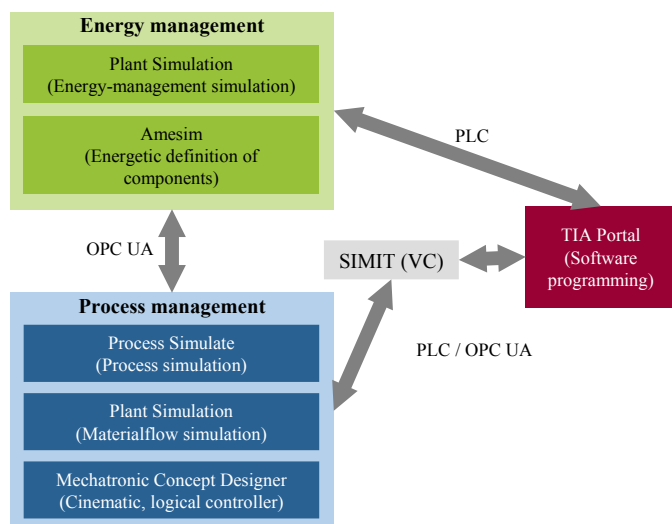


Fig.4. Overview of possible tools used for VC regarding energy consumption.

3.3. VC in SMC

The field of SMC is characterized by high-quality unique projects. In most of the cases, those projects force a so-called “First Time Right”, which means that there is no time for many iterations in the engineering process so that the engineering results have to work at the first attempt. Therefore, the application of VC points out a high potential to prevent errors in the engineering process as well as in the commissioning. The integration of the modelling process in the engineering allows the engineering team to work on corporate basis across domains. Due to the early cooperation, potential discrepancies can be found easily. Fig. 5 shows the different engineering phases as well as the buildup of the DT of the production system. After the first iteration in the modelling process (1,2) the focus is on the early validation of the PLC code and the

kinematics of the production system (3,4). In iteration 5 the constructive changes resulting from real commissioning have to be transferred to the behavior model. Herewith, the basis for the DT of the production system in serial production is established.

The field of activity of the SMC covers besides engineering and commissioning, the run-up support as well as trainings. Here, the simulation model with the validated PLC code can be used in trainings and for maintenance scenarios (6). Concerning the service, software updates can be reviewed in advance of the maintenance layer to save valuable production time.

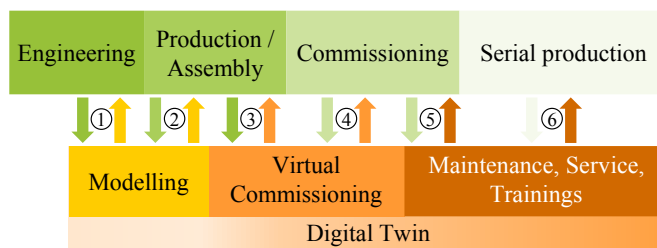


Fig. 5. Development of the DT using VC in SMC.

Generally, an apposition of special machines results in a production system. The latter, which can be provided by different suppliers, makes it difficult to implement a single model for the VC of the entire system. Hence, a strict guideline has to be developed in terms of the mechanical and electrical construction as well as the software tools to guarantee an appropriate usage of VC. Furthermore, SMCs build their machines mostly once for a single customer. Thus, once created simulation models cannot be reutilized without more effort, whereby parametrizable models play an important role.

4. VC as the cornerstone of the DT

In general, a DT is the digital representation of things from the real world [47]. Going more into detail, there are different definitions of DTs dependent of the point of view. A detailed comparison of DT definitions is given in [48]. Considering the production system, there are three different DTs. Fig. 6 shows those as well as their intersections.

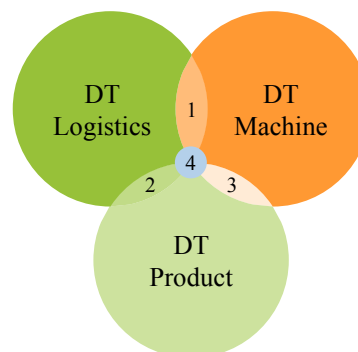


Fig. 6. Interaction of the DTs: logistics, machine and product.

Regarding the VC of production systems, the created simulation model is the basis for its DT. In the next step, the

model needs to be connected to the real machine, enabling a real-time virtual representation of the productive system. In doing so, it is easier to retrace production downtimes caused by the machine and avoid them in future as real-world failures can be re-enacted and analyzed on the virtual image.

Besides the DT of the machine, there is the DT of the product. It consists of all data concerning its individual production as well as a behavior model. Using the DT of the product, information with regard to the product life can be gained and potential malfunctions can be detected. Just like the real product, its DT is linked with the DT of the machine by the manufacturing process, marked as (3) in Fig. 6. The manufacturing processes represent the interfaces between the DTs of machines and products.

At last, logistic processes of the production system periphery are modelled digitally. The DT of the logistics enables a proactive production planning and a focused analysis in case of occurring bottlenecks and other problems in the material flow. It also enables a current system state-based prediction of future system behavior and performance in different scenarios, allowing for a predictive optimization of controllable system parameters such as order sequence. The interactions between the logistic processes and the machine (1) as well as the product (2) are also mapped digitally to benefit from correlations and potential synergies. A possible implementation of a DT of logistic processes is a material flow simulation model, which is linked to the real logistic processes. [49] Regarding the logistics and the machines, DTs can base on the simulation models from early planning stages enriched by additional information from serial production.

The intersection of all three DTs (4) is not relevant for most simulation applications due to the different points of view. However, the interchange of relevant data between the different DTs is necessary as mentioned before.

5. Conclusion and outlook

This paper shows that the focus in VC is currently on checking the automation solution in form of the PLC code. It is only on ensuring a functioning process control of the industrial plant. Safety or energy technology are left out and are in most of the cases not simulated. The greatest challenge in this context is the efficient creation of models. Hence, it is necessary for component manufacturers to provide extended digital representations of their products to their customers, exceeding the predominant supply of construction data. Finally, the current research focus regarding the approaches for VC and the behavior model abstraction are given.

Nevertheless, it is obvious that the models being necessary for VC have not served their purpose after successful commissioning of the industrial plant, but they should accompany the industrial plant as a DT beyond the product development process. However, until this is ready for series production, various challenges have to be solved.

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